

Summary:

Electrical and Computer Engineering graduate with a strong academic background in VLSI systems, microelectronics, digital design, and hardware development. Hands-on experience in complex engineering projects involving circuit design, simulation, validation, and system analysis. Strong analytical and problem-solving skills with a passion for semiconductor technologies and multidisciplinary hardware engineering environments.

Education:

- **2020–2024: The Hebrew University of Jerusalem , B.Sc. in Electrical and Computer Engineering**
 - Focus: Microelectronics, VLSI Design, and Digital System
 - Core Courses: The Physics of Semiconductor Devices, Advanced VLSI, Interfacing Analog and Digital Worlds (Verilog), Analog Electronics Lab, Python, Data Structures, Micro-Electronics Devices Lab, Engineering Project Workshop.

Certificates :

- **2026 : AI Engineer Career Accelerator Program**
Led by Mohammad Kabajah, Sr. AI/Cloud Infrastructure Architect at AWS
 - Developing an **AI-powered recruitment platform** for CV analysis, candidate-job matching, gap identification, and tailored recommendations.
 - Applying **LLMs, agentic AI concepts, document processing, information retrieval, and AI-assisted development** within a team-based engineering project.
- **2025: NVIDIA Deep Learning Institute (DLI)**
 - Accelerate Data Science Workflows with Zero Code Changes
 - An Even Easier Introduction to CUDA.
- **2024: Tel Aviv University**
 - Business Economics
 - People and Systems Management (Micro & Macro)
 - Statistical and Analytical Models for Management
- **2023: Bridgz Company** - Three-month theoretical and practical training programs focused on chip design and RTL-to-GDSII flow :
 - Industry -Oriented Training in VLSI (Physical Design)
 - Chip Design Verification
- **2022: Technion Israel Institute of Technology**
 - All-Digital Phase-Locked Loops Course

Final Projects:

- **Two-Step Time-Domain ADC with Delay-Tracking SAR TDC –**
 - Designed and simulated a **6-bit Time-Domain ADC** in **14nm CMOS** (Cadence Virtuoso)
 - Implemented Sample & Hold, VTC, and Delay-Tracking SAR TDC blocks
 - Performed transient and corner simulations, investigated timing behavior, and validated performance against design specifications.
 - Achieved **1.25 GS/s, 7.4 mW, 50 dB SNDR**
- **MEMS-Based Optical System –**
 - Designed and analyzed an electromagnetically actuated **MEMS micromirror** for high-speed beam steering.
 - Developed electro-mechanical models using COMSOL and MATLAB and validated simulation results against analytical calculations.
 - Achieved $\pm 10^\circ$ tilt, >100 kHz resonance, and <10 μ s response time

Experience:

- **2020–2023: Tutor**, The Hebrew University of Jerusalem – Tutored courses in :
 - Classical mechanics
 - Analog electronics, and digital electronics.
- **2020–2022: Shift Manager**, Bloomfield Science Museum – Delivered interactive science workshops, explained scientific concepts through a wide range of museum exhibits, trained museum guides, and supervised daily shift operations.

Technical Skills:

Digital Design & Verification: Verilog / SystemVerilog, RTL simulation, debugging & root-cause analysis , Functional validation, test, scenario development, waveform analysis ,Understanding of verification flows, correctness checking, and ASIC design concepts

VLSI & Systems: RTL-to-GDSII flow understanding, digital/analog circuit fundamentals ,SoC architecture concepts, power-efficient design considerations ,Microelectronics & semiconductor device fundamentals

Programming & Scripting: Python, C++, MATLAB, C, C#, SystemVerilog

Hardware & Lab Tools: Cadence Virtuoso , COMSOL Multiphysics , Oscilloscopes, logic analyzers, spectrum analyzers, signal generators

Community Involvement:

- **2022: Educational Assistant, Shams Association** – Facilitated cognitive intelligence games for children, observed and documented their behaviour, and supported personalized learning strategies.
- **2021–2022: Chess Instructor, Chess Association** – Taught structured chess lessons to enhance students’ strategic thinking, focus, and problem solving skills.

Languages:

| Hebrew – Fluent | English – Fluent | Arabic – Native |